

AMAZON FRESH SALES ANALYSIS

SQL MINI PROJECT REPORT

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1. Abstract

The Amazon Fresh Sales Analysis project was developed to analyse business performance using MySQL and Microsoft Power BI. SQL was used to organize, query, and analyse the data, while Power BI was used to create an interactive dashboard with key business insights. The dashboard provides a clear overview of sales, customers, products, and regional performance, enabling better business decision-making.

2. Problem Statement

Amazon Fresh is an e-commerce platform specializing in groceries and daily essentials. As the platform expands, its management seeks to optimize operations, enhance customer satisfaction, and boost revenue. The challenge lies in effectively managing large volumes of data, including customer profiles, product inventories, supplier information, orders, and customer reviews.

The problem is to design a relational database for Amazon Fresh that organizes this data efficiently, allows for meaningful analysis, and supports critical business decisions. This project aims to solve key business problems, such as identifying top-performing products, analyzing customer purchasing patterns, and ensuring inventory management aligns with demand trends.

3. Objectives

- ✓ To design and implement a relational database for the Amazon Fresh dataset using MySQL.
- ✓ To perform SQL operations such as data retrieval, filtering, joins, aggregations, and subqueries for business analysis.
- ✓ To analyse customer, product, sales, and regional performance using SQL queries.
- ✓ To develop an interactive Power BI dashboard for visualizing key business metrics.
- ✓ To generate actionable business insights that support data-driven decision-making.
- ✓ To enhance analytical and business intelligence skills through an end-to-end data analytics project.

4. Dataset Description

The Amazon Fresh dataset consists of multiple interconnected tables representing different business entities involved in the e-commerce platform. The dataset includes information related to customers, products, suppliers, stores, orders, order items, and customer reviews. Each table is connected through primary and foreign keys, forming a relational database structure that supports efficient data management and analysis.

The dataset was imported into MySQL Workbench and used for SQL-based analysis. After completing the SQL implementation, the processed data was connected to Microsoft Power BI to create interactive dashboards and generate meaningful business insights.

Tables Used

1. Customers
2. Products
3. Orders
4. Order Items
5. Suppliers
6. Stores
7. Reviews

5. Tools and Technologies Used

S.No	Tool	Purpose
1	MySQL Workbench	Database creation, data management, and SQL query execution
2	Microsoft PowerBI	Data visualization and interactive dashboard development
3	Microsoft Excel	Data verification and preliminary data review

6. Project Methodology

The Amazon Fresh Sales Analysis project was carried out using a structured approach, beginning with database creation and ending with interactive dashboard development. The complete workflow is outlined below:

Step 1: Dataset Collection and Understanding

- ✓ Collected the Amazon Fresh dataset containing customer, product, supplier, store, order, order item, and review information.
- ✓ Understood the relationship between different tables and identified the relevant attributes required for analysis.

Step 2: Database Creation

- ✓ Imported the dataset into MySQL Workbench.
- ✓ Organized the data into relational tables.
- ✓ Identified and established primary key and foreign key relationships to maintain data integrity.

Step 3: SQL Analysis

- ✓ Executed SQL queries to retrieve, filter, update, and analyze data.
- ✓ Performed joins, aggregations, grouping, sorting, subqueries, and normalization to answer business-related questions.

Step 4: Power BI Dashboard Development

- ✓ Connected the SQL database to Microsoft Power BI.
- ✓ Created an interactive data model and developed multiple dashboard pages using KPI cards, charts, slicers, and navigation buttons.

Step 5: Business Insights

- ✓ Analyzed customer behavior, product performance, regional sales, revenue contribution, and overall business performance.
- ✓ Generated meaningful insights and recommendations to support business decision-making.

7. Implementation

7.1. Data Preparation (Excel)

The Amazon Fresh dataset was initially reviewed in Microsoft Excel to understand its structure and verify the available data. The dataset consisted of multiple files representing different business entities, including customers, products, suppliers, stores, orders, order items, and customer reviews. Before importing the data into MySQL, the dataset was carefully examined to ensure that the records were complete and organized. Basic validation was performed to identify inconsistencies, remove duplicates, verify column names, and understand the relationships between different tables. This initial review helped establish a clear understanding of the dataset and prepared it for database implementation.

7.2. Database Implementation (MySQL Workbench)

7.2.1. Database Design and Data Modeling

The Amazon Fresh dataset was imported into MySQL Workbench to create a structured relational database. The database was designed by organizing the dataset into multiple tables representing different business entities such as Customers, Products, Orders, Order Items, Suppliers, Stores, and Reviews. Each table was assigned appropriate primary keys and foreign keys to establish relationships

and maintain referential integrity. An Entity Relationship (ER) Diagram was created to visualize the relationships between the tables. This data model served as the foundation for executing SQL queries and performing business analysis efficiently.

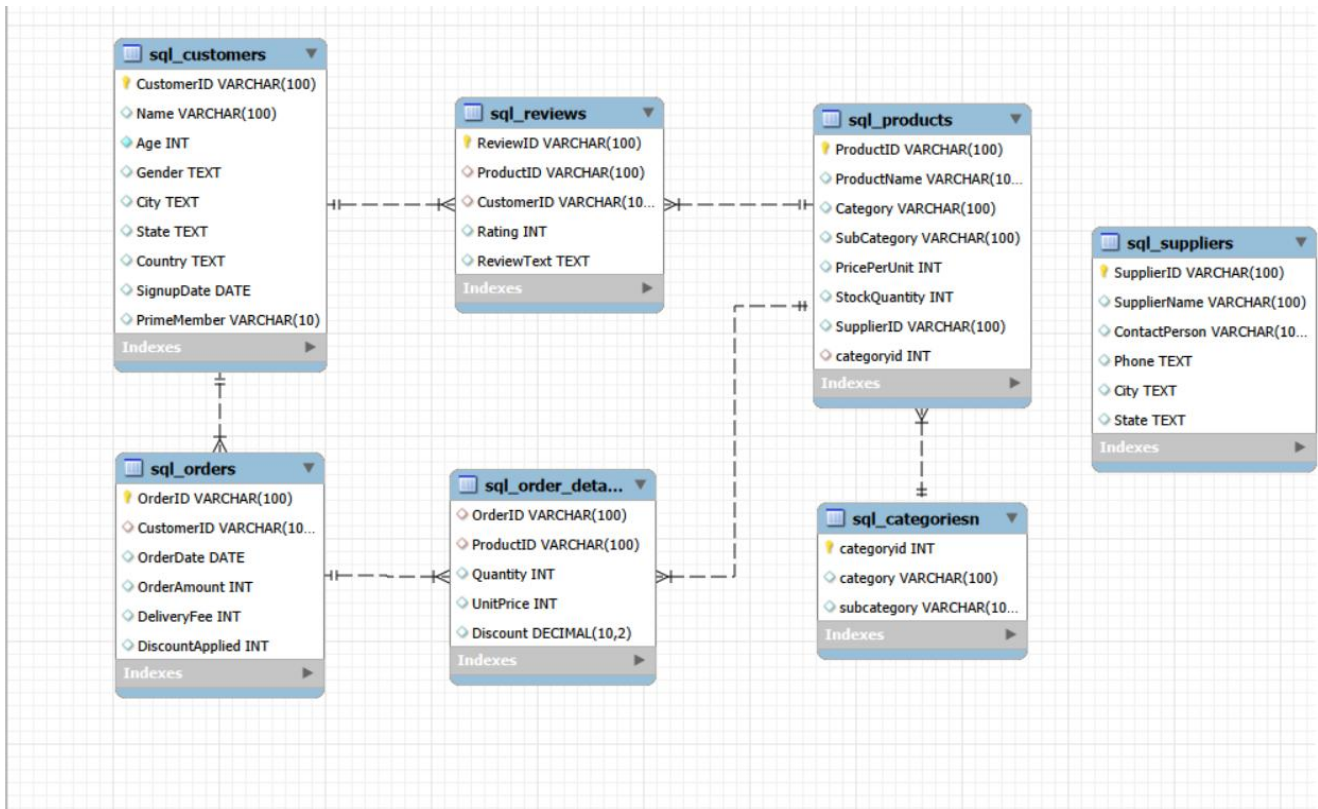


Fig.1. Entity Relationship (ER) Diagram of the Amazon Fresh Database

7.2.2. Table Creation and Constraints

The database tables were created using SQL Data Definition Language (DDL) commands. Appropriate constraints such as PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, CHECK, and DEFAULT were implemented to ensure data consistency and integrity. These constraints helped prevent invalid data entry and maintained reliable relationships between the tables.

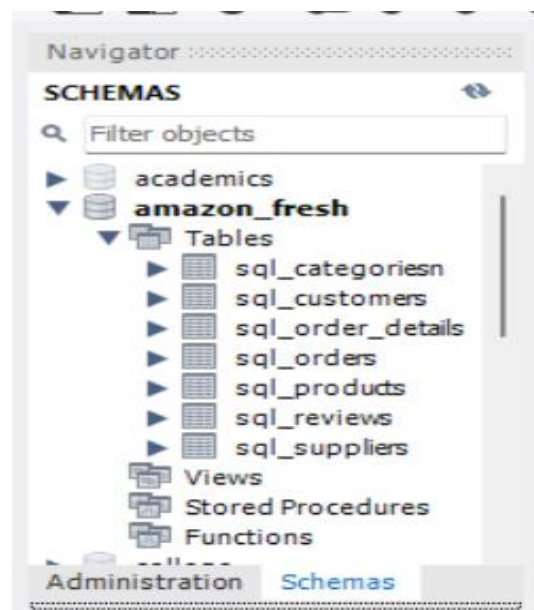


Fig.2. Tables created in MySQL Workbench

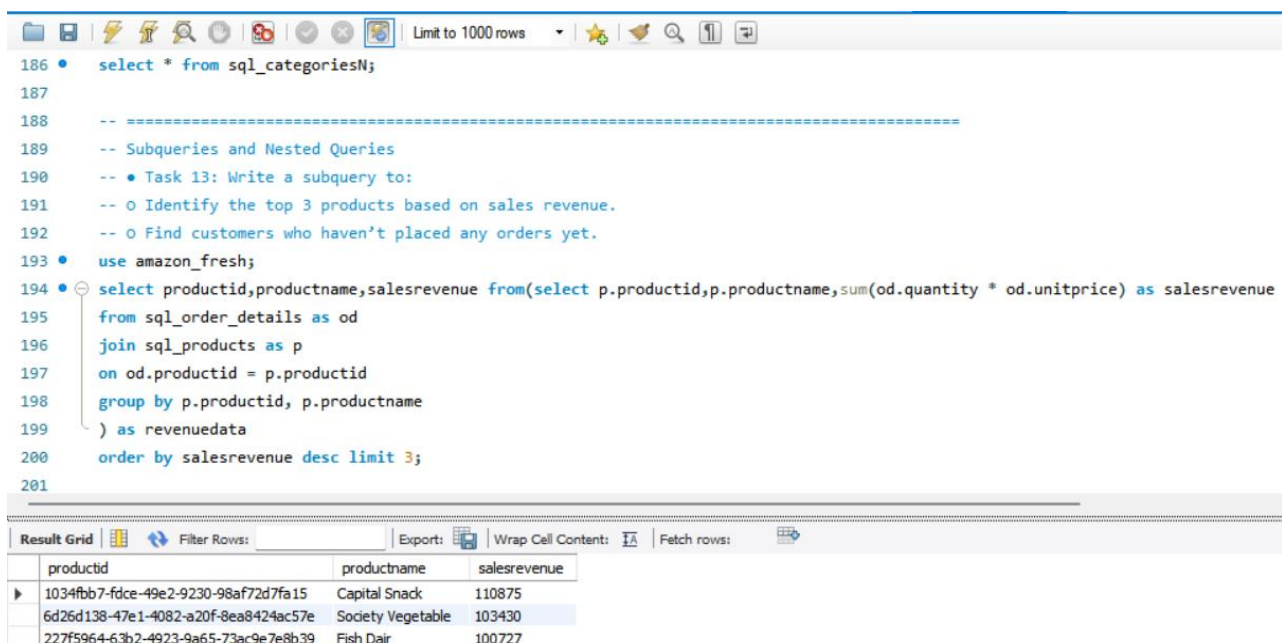
7.2.3. Data Manipulation

SQL Data Manipulation Language (DML) operations were performed to modify and maintain the database records. Operations such as INSERT, UPDATE, and DELETE were used to add new records, update existing information, and remove unnecessary data based on the project requirements. These operations ensured that the database remained accurate and up to date throughout the analysis process.

7.2.4. Data Analysis Using SQL

The following SQL operations were performed to analyze the Amazon Fresh dataset:

- ✓ Retrieved and filtered records using SELECT and WHERE clauses. Grouped and summarized data using GROUP BY and HAVING clauses. Sorted query results using the ORDER BY clause.
- ✓ Calculated business metrics using aggregate functions such as SUM(), COUNT(), AVG(), MIN(), and MAX(). Analyzed customer, product, and sales data to generate meaningful business insights.



The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, a search icon, and a 'Limit to 1000 rows' dropdown. The SQL editor contains the following code:

```
186 • select * from sql_categoriesN;
187
188 -- =====
189 -- Subqueries and Nested Queries
190 -- • Task 13: Write a subquery to:
191 -- o Identify the top 3 products based on sales revenue.
192 -- o Find customers who haven't placed any orders yet.
193 • use amazon_fresh;
194 • select productid,productname,salesrevenue from(select p.productid,p.productname,sum(od.quantity * od.unitprice) as salesrevenue
195 from sql_order_details as od
196 join sql_products as p
197 on od.productid = p.productid
198 group by p.productid, p.productname
199 ) as revenuedata
200 order by salesrevenue desc limit 3;
201
```

Below the editor is the 'Result Grid' tab, which displays the query results in a table:

productid	productname	salesrevenue
1034fbb7-fdce-49e2-9230-98af72d7fa15	Capital Snack	110875
6d26d138-47e1-4082-a20f-8ea8424ac57e	Society Vegetable	103430
227f5964-63b2-4923-9a65-73ac9e7e8b39	Fish Dair	100727

Fig.3: SQL Query and Result for Top Three Products by Sales Revenue

7.2.5. Business Analysis Using SQL

The following business analyses were performed using SQL queries:

- ✓ Calculated the total revenue generated for each order.
- ✓ Identified high-value customers based on their total spending.
- ✓ Analyzed customer purchasing patterns and order history.
- ✓ Evaluated product performance across different categories.
- ✓ Identified suppliers with the highest product availability.
- ✓ Retrieved the top-performing products based on sales revenue.
- ✓ Identified customers who had not placed any orders.
- ✓ Applied normalization techniques to improve database structure and maintain data integrity.

7.2.6. SQL Implementation Outcome

The SQL implementation successfully achieved the following outcomes:

- ✓ Developed a well-structured relational database.
- ✓ Performed efficient data retrieval and analysis using SQL queries.
- ✓ Generated meaningful business insights from customer, product, and sales data.
- ✓ Prepared the processed data for visualization in Microsoft Power BI.
- ✓ Established a strong foundation for interactive dashboard development.

7.3. Dashboard Development (Microsoft Power BI)

7.3.1. Data Integration

After completing the SQL analysis, the processed data was connected to Microsoft Power BI for visualization. The imported tables were verified, and the relationships between the tables were established to create an efficient data model for analysis.

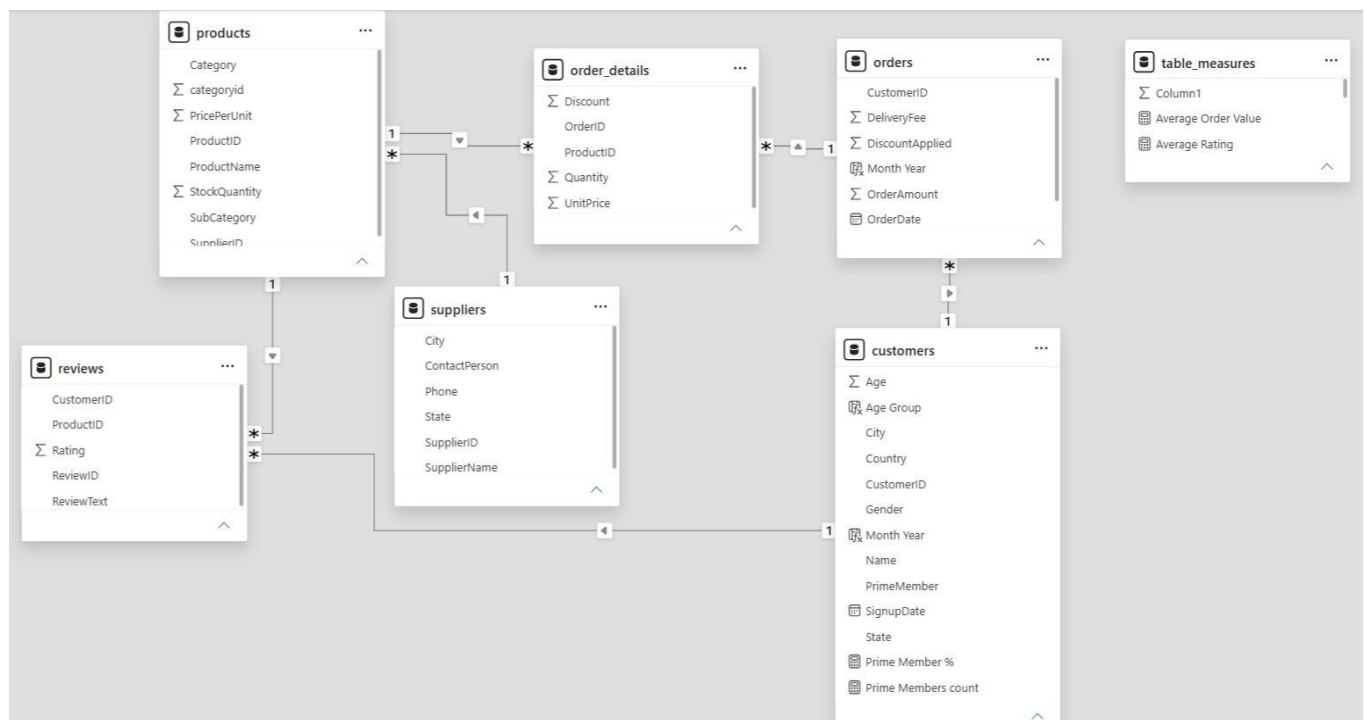


Fig.4: Power BI Data Model Showing Relationships Between the Amazon Fresh Tables

7.3.2. Dashboard Design

The Power BI dashboard was designed to provide a comprehensive overview of Amazon Fresh's business performance. The dashboard includes multiple report pages, each focusing on a specific area of business analysis. The dashboard consists of the following pages:

- ✓ Home Page
- ✓ Business Overview
- ✓ Product Insights
- ✓ Customer Insights
- ✓ Customer Details
- ✓ Performance Insights



Fig.5. Home Page Design

7.3.3. Data Visualization

Various visualizations were created to analyze different aspects of the business, including:

- ✓ Category-wise Sales Analysis
- ✓ Product Performance Analysis
- ✓ Top 10 Best-Selling Products
- ✓ Customer Distribution Analysis
- ✓ Top Customers by Revenue
- ✓ State-wise Sales Performance
- ✓ Revenue Contribution Analysis
- ✓ Discount Performance Analysis

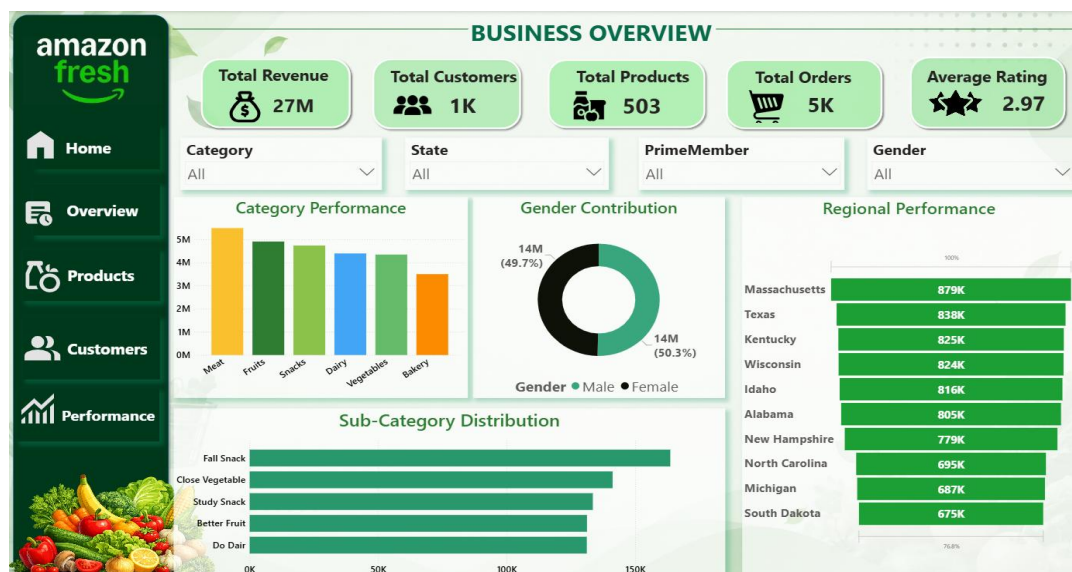


Fig.6. Business Overview Page showing Key Performance Indicators

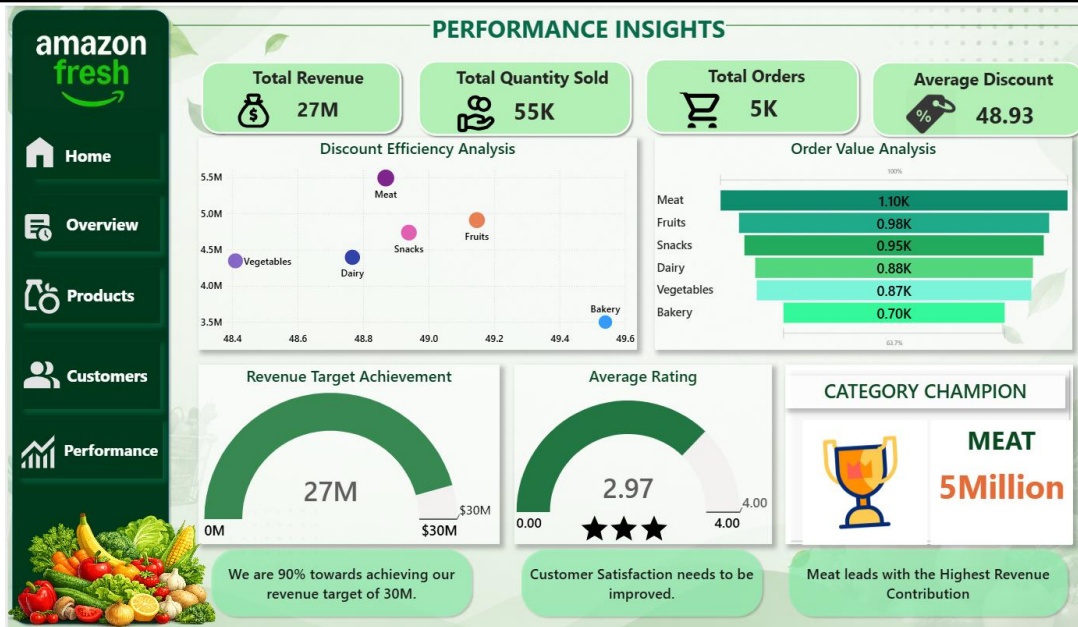


Fig.7. Business Performance Dashboard

7.3.4. Interactive Features

To improve user experience and dashboard interactivity, the following features were implemented:

- ✓ Slicers for dynamic data filtering.
- ✓ Navigation buttons for seamless page transitions.
- ✓ Cross-filtering between visuals.
- ✓ Consistent Amazon Fresh dashboard theme.
- ✓ Professional formatting of charts, KPI cards, and report pages.

7.3.5. Dashboard Outcome

The Power BI dashboard successfully transformed the SQL analysis into an interactive business intelligence solution. It enabled users to monitor key performance indicators, analyze sales trends, evaluate customer and product performance, and support data-driven business decisions through intuitive visualizations.

8. Business Insights

The analysis and dashboard provided the following key business insights:

- ✓ Identified the highest revenue-generating product category.
- ✓ Determined the top-performing state based on total sales.
- ✓ Identified the best-selling products based on revenue.
- ✓ Recognized high-value customers through purchase analysis.
- ✓ Evaluated customer distribution across different states and gender.
- ✓ Analyzed the impact of discounts on revenue generation.
- ✓ Monitored overall business performance using KPI metrics.

- ✓ Enabled better decision-making through interactive visualizations.

9. Challenges Faced

During the implementation of this project, a few challenges were faced :

- ✓ Understanding the relationships between multiple tables in the dataset.
- ✓ Designing an efficient relational database structure.
- ✓ Applying SQL constraints while maintaining data integrity.
- ✓ Writing complex SQL queries involving multiple table joins.
- ✓ Creating meaningful business insights from raw data.
- ✓ Designing an interactive and user-friendly Power BI dashboard.
- ✓ Maintaining consistency in dashboard layout, formatting, and navigation.

10. Conclusion

The Amazon Fresh Sales Analysis project successfully demonstrated the complete data analytics workflow, from database creation and SQL analysis to interactive dashboard development in Power BI. The project transformed raw business data into meaningful insights that support informed decision-making. Overall, this project strengthened practical knowledge of SQL, database management, and business intelligence while providing valuable hands-on experience in solving real-world business problems.